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This is a proposal for pre-flight ground-based research to explore emerging new techniques in coherent optical frequency synthesis and measurement, generation of precise atomic clock signals based on ultrahigh Q optical transitions, and distribution of stable optical clocks across communication networks, including free-space propagation. Our work envisions fundamental atomic clock experiments to be performed in space, and likely within the context of an actively-manned space station.
The proposed activities will have impacts to a number of important scientific and technological thrusts of NASA, including spaceborne optical interferometry, enhanced formation flying, and gravitational wave detection. Our main objective is to use ultra-stable lasers in space-based fundamental research, in particular for the gravitational and relativistic physics that has become more urgently relevant with LISA. Many general relativity experiments such as LISA and The Laser Interplanetary Ranging Experiment (LIRE) will need an unprecedented level of laser stability to

carry out successful missions. The proposed work should also make a strong impact to the cold atom research programs through the enhanced control of matter-field interaction.

Task Description:

We will study physical limitations of stable optical frequency sources such as diode-pumped solid state lasers stabilized by a combination of molecular and cavity resonances. We will employ our new understandings in the modulation process that are critical for signal recovery and the resultant system accuracy. Another concrete set of subtasks concerns active suppression of laboratory vibrations that affect all high-resolution measurements. Low-noise sensing, orthogonalization of the several vibrational degrees of freedom, and design of appropriate servo controllers are all part of this sub-task. Basically all measurements encounter environmental limitations as the measurement precision is enhanced. These thermal and acceleration-induced changes may occur naturally on longer timescales for experiments associated with a space-based research environment, particularly in a microgravity environment. We will also actively pursue a cold atom based optical frequency standard, building upon our recent advances in understandings of the Alkaline earth Sr atoms that offers a particularly attractive optical transition with potential precision and accuracy rivaling single-ion based systems. Once a laser system with an unprecedented stability is developed, we will transfer the stability of this optical oscillator to a wide range of desired optical and microwave spectra, using our recently developed methodology of establishing tight phase-coherent links among optical oscillators and between optical and microwave oscillators using femtosecond-laser-based optical comb technology. One concrete plan is to explore a portable fiber-laser-based system that will be self-referenced to produce stabilized optical and RF clock signals that can be used simultaneously for laser ranging and time keeping in spaceborne experiments. The significance of this development will also be reflected in the research area of space based laser cooling and atomic physics within NASA's microgravity fundamental physics program, for instance, in the production of ultracold atoms and in the study of coherent control and manipulation of atomic wavepackets. Finally, during the course of study we will pay close attentions to practical issues of transferring these ground-based technologies to adapt to the operating environment of spacecraft.

The most important milestones are: • Precision spectroscopy of super-narrow intercombination lines in ultra-cold Sr atoms. Explore high quality molecular transitions in cavity-enhanced configurations for portability. Build optical frequency standards with stability, precision and accuracy approaching and potentially surpassing the best current systems. • Accurate control of the modulation and signal recovery processes; Stability of cw optical local oscillator approaching 1 * 10-15 at 1 s. •

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

We have made significant progress in all fronts of the proposed research topics since our proposal was funded in April, 2004. During this interval we continued our exploration of the remarkable capability we have developed in connecting optical and radio frequency domains in a phase-coherent manner. Femtosecond lasers have been stabilized to produce exceptional performances in both short-term (in terms of individual optical comb linewidth) and long-term (in terms of optical frequency reproducibility and accuracy) time scales. A new type of resonant photodetector is developed at the 10 GHz frequency range that enabled noise reduction in the conversion process from optical to microwave frequencies. We have developed new types of mode-locked femtosecond lasers that have simplified the implementation of optical clockwork. We recently demonstrated an optical clock based on vibrational transitions in the methane molecule. With the simple new method, the methane molecule forms the "pendulum" of the clock, but oscillates 88 trillion times per second, a million times too rapidly to count. A single femtosecond laser can be used to create a microwave signal with all the stability and accuracy of the methane infrared transition. The femtosecond laser can be thought of as the "clockwork," or the gearing that allows the rapid optical oscillations to be transferred to much slower microwave oscillations which can then be counted as the clock signal. Prior to this work, optical clockwork requires stabilization of both the repetition and the carrier-envelope offset frequencies associated with the femtosecond laser. In the current approach, the femtosecond laser's output contains two optical spectral features with their frequency difference corresponding to the infrared transition frequency of the methane molecule. In doing so one cancels the original laser's fluctuating carrier-envelope offset frequency in the resulting optical comb in IR. In other words, the IR optical comb contains tens of thousands optical frequency markers that are exactly integer multiples of the laser's original repetition frequency. This allows a direct phase-coherent connection between the optical transition frequency and the repetition frequency in the microwave domain. This approach can be used to cover a broad range of infrared frequencies and it is expected to be more broadly applicable to making very precise measurements of many molecular transitions, allowing detailed study of molecular structure and dynamics.

With the advent of optical atomic clocks and the associated superior short-term frequency stability, transfer of signals linked to such frequency standards over an appreciable distance with a minimal loss of stability is an important subject for research. In previous work we have studied the transfer of the optical carrier of a cw laser over an installed fiber network. A femtosecond laser located at the remote end can be phase locked to the incoming cw laser carrier, redistributing the optical frequency to the microwave domain via the mode-locked laser's repetition frequency. We have now demonstrated high-stability transfer of optical frequency standards using mode-locked laser sources that are transmitted through this fiber network, which offers several advantages over the transfer of the optical carrier of a cw laser. When the mode-locked laser is phase coherently connected to an optical atomic clock, both the optical carrier frequency and the repetition rate of the mode-locked laser are connected to a single optical frequency standard. Therefore, a single fiber link allows a user at the remote end to either simply use a fast photodiode to recover the radio frequency reference signal or establish a more elaborate, direct optical carrier connection. Simultaneous existence of both the optical phase and the pulse repetition rate information would also enable novel measurement capabilities on material dispersion and length determination.

We have confirmed that the instability of the transfer process for the repetition frequency over a fiber network is nearly the same as that of optical carrier transfer of a cw laser, and is almost an order of magnitude better than that of transfer of a modulated carrier, which is an alternate approach to transferring a radio frequency reference. Furthermore, it is clear that a pulsed transfer process preserves the stability of the optical carrier as in the cw optical transfer process. In addition, the interplay between the mode-locking conditions of the mode-locked source and the transfer process are explored, and optimum conditions are found to minimize the instability introduced by the transfer and subsequent detection. For the implementation of frequency transfer over long distance fiber networks, it is necessary to include

Task Progress:

erbium-doped fiber amplifiers in the network. The effects of such an amplifier on the instability of the transfer of the repetition frequency are also studied, and the results show the instability remains well below that of modulated-carrier transfer. The transfer of both the repetition frequency and optical carrier will enable the active control of both the group-delay and phase-delay of the transfer path in the future, which will make possible time-domain experiments such as remote synchronization of two mode-locked lasers.

In taking this work a step further, we have also recently demonstrated several important advances to pulsed signal transfer over a fiber link that are vital for both precision frequency metrology and highly stable timing synchronization. The first important advance is control of the dispersion of the fiber link, resulting in short pulses after transmission that are necessary to eliminate the instability we found to be caused by the photodetectors. Second, we have implemented fast, active cancellation of the fiber transmission noise. This reduces the instability of transfer to the measurement noise floor for averaging times greater than 1 s, corresponding to instability of 7E-15 at 1 s. Third, we have completed studies of noise occurring on faster timescales that will be crucial for potential timing-synchronization applications. The active noise cancellation has reduced the residual timing jitter, integrated from 1 Hz to 100 kHz, to 37 fs for an installed fiber network and 20 fs for a laboratory-based network. To our knowledge, these numbers represent the lowest reported jitter for transfer of a timing signal over km-scale distances. These results should be immediately applicable to several research areas that necessitate the transfer of an ultralow-jitter timing signal, such as synchronization throughout a linear-accelerator facility and long-baseline astronomical observations requiring coherent data collection among remotely located telescopes.

Along with this development on ultrastable and precise signal transfer, we have developed two synchronized and phase-stabilized mode-locked fiber lasers. They are the workhorse for the signal transfer project but they also represent the future system of a portable femtosecond laser system that will enable simultaneous time-keeping (optical atomic clock) and distance ranging (see the next paragraph) tasks.

Based on these developments, we have proposed a novel and practical scheme for high precision, absolute length measurement for an arbitrary optical distance of a few meters to beyond 1E6 meters. The approach takes advantage of the remarkable technology advances in phase stabilized femtosecond lasers. An ultrafast pulse train allows a determination of absolute distance by using the time-of-flight information from incoherent measurement approach. Meanwhile, a phase-stabilized femtosecond pulse train would also allow one to build optical interferences fringes when pulses traversing different arms of an interferometer are made to overlap and interfere, given a proper adjustment of the parameters associated with the mode-locked laser. Essentially we can enable a “white-light” interferometry condition to occur at any desired length difference between the two arms in the interferometer. Such a combined measurement capability allows an optical wavelength resolution to be achieved for absolute length measurement over a large dynamic range. This proposal has attracted interest from a number of US industrial firms and national laboratories for actual laboratory demonstration and system implementation.

One of the most noteworthy developments in the past year in the laser stabilization community is our innovative design of a novel intra-vacuum suspension system for passive optical cavities which offers considerable isolation from high frequency mechanical vibrations along with an even more benign and stress-free support. Narrow linewidth lasers are necessary as local oscillators in optical frequency standards and serve as precision spectroscopic probes and frequency/phase-transfer flywheels. A free-running laser is most often servo-locked to the resonance of a Fabry-Perot cavity: the remaining instability of the locked laser is a combination of the instability of the length of the frequency reference itself and the defects of the servo-locking system. Environmental noise sources that modulate the cavity length are mainly of acoustic origin in the range above 50 Hz and seismic origin from 1 to 50 Hz. For Fourier frequencies less than 1 Hz, temperature changes affect the cavity length as well as other dimensions in the experiment that may inadvertently contribute to the instability.Successful solutions to cavity length changes caused by acoustic and seismic noise have been based on low-pass filtering and active vibration cancellation systems. We have demonstrated that designing for low acceleration sensitivity of the cavity length is a great help. For this purpose, shorter cavities are better since the vibrational sensitivity is proportional to the cavity length. In our experimental geometry we arrange the cavity support points to be in the symmetry plane between the two mirrors on the cavity ends. For acceleration along the cavity axis, each half of the cavity is deformed on either side of the central mounting plane. The decrease in length on one side of the mounting is mostly compensated for by an increase in length on the other. This design for a cavity support system is simple, robust, and offers reduced sensitivity to the cavity’s length changes caused by acceleration. The vibration isolation requirements for a stated performance are significantly reduced, and few, if any, disadvantages are introduced. With this suspension system for a passive optical cavity used for laser frequency stabilization, we experimentally demonstrated effective reduction of the vibration sensitivity of the cavity length. The work has led to a simple and compact reference cavity system for laser stabilization at the level of 1-Hz linewidth.

On a parallel front we have achieved direct stabilization of a femtosecond laser based optical frequency comb by a high-finesse passive optical cavity. Improving the precision in which the repetition rate and carrier-envelope phase of ultrashort pulses can be controlled has obvious and important consequences for many scientific applications. The additional complexity arising from simultaneous stabilization of the two degrees of freedom of a femtosecond laser requires thorough considerations of the control loop designs and the restrictions due to dispersive phase shifts inside the reference cavity. The broader goal of this work is to not only elevate the stability of ultrafast lasers to levels comparable with state-of-art CW lasers, but also enable a number of applications based on external ``fs enhancement cavities" to provide attractive field-enhancement effects and sensitive detection capabilities in nonlinear optics and atomic and molecular spectroscopy. We have performed detailed comparisons of two distinct stabilization schemes, leading to new understanding of the optimum conditions for cavity stabilization and limitations on the ability to transfer the frequency stability of the cavity to the microwave domain. The stability of the frequency comb is explored in both the optical and the radio frequency domain. With an independent, stable CW laser, we verify that the linewidth and stability of the optical comb components, respectively, reach below 300 Hz and 5E-14 at 1 s averaging time, both limited by the reference CW laser.

The progress we have made on the proposed precision spectroscopy of super-narrow intercombination lines in ultra-cold Sr atoms has been rapid and multi-facet. Neutral-atom-based optical frequency standards have become serious contenders for the realization of highly stable and accurate optical atomic clocks. Progress in this field is rapid, with significant advances in experimental techniques from several related fields including laser cooling to ultracold temperatures, trapping configurations suitable for highly accurate clocks, the development of high quality optical local oscillators, and precise frequency measurement and distribution enabled by femtosecond optical combs. We are pursuing a system based on intercombination lines in ultracold Sr atoms localized spatially in a Lamb-Dicke regime while the trapping potential is designed to minimize its influence to the clock transition frequency. This scheme is being

implemented using a far-off-resonance dipole trap operating at a wavelength where the ground and the excited state of the clock transition experience exactly the same AC Stark shifts. States with scalar polarizabilities are preferred, to avoid the problem of the complex and sometimes uncontrolled light polarization inside an optical lattice. The fermionic strontium isotope, 87Sr, offers a nearly satisfactory solution. The nuclear magnetic dipole moment of 87Sr makes the 5s2; 1S0 – 5s5p; 3P0 transition weakly dipole allowed, with a predicted linewidth of about 1 mHz. The two clock states have very small electronic angular momenta (due to hyperfine mixing) and their polarizabilities are nearly scalar. However, the hyperfine structure can cause a clock frequency shift (possibly as large as 10 Hz) by fluctuations of the lattice light polarization. The large nuclear spin (I = 9/2) also brings complexity in state preparation and field control.

We have developed an improved optical atomic clock scheme that utilizes two lasers to establish coherent coupling between the 5s2; 1S0 ground state of 88Sr and the first excited state, 5s5p; 3P0. 88Sr provides clock states with a true scalar character which avoids the complications associated with 87Sr. The coupling is mediated by the broad 5s5p; 1P1 state, exploiting the phenomenon of electromagnetically induced transparency. The effective linewidth of the clock transition can be chosen at will by adjusting the laser intensity. By trapping the 88Sr atoms in an optical lattice, long interaction times with the two lasers are ensured; Doppler and recoil effects are eliminated. Based on a careful analysis of systematic errors, a clock accuracy of better than 2 x 10^{-17} is expected.

We have performed precision spectroscopy of ultracold 88Sr in free space. With reference to the Cs primary standard and with a detailed investigation and analysis of systematic frequency shifts, we determine the absolute frequency of the 1S0 - 3P0 clock transition with a statistical (systematic) uncertainty of 20 (26) Hz. This level of measurement precision represents an improvement of more than 200 times over recent studies and represents currently the most accurate measurements in a neutral Sr system. Collision-related frequency shifts can significantly impact both the stability and accuracy of microwave and optical frequency standards. These effects have been studied in detail for both Cs and Rb microwave clocks. For optical frequency standards based on intercombination transitions in alkaline earth atoms, it is expected that the dependence of the fractional frequency shift on the atomic density is reduced by more than three orders of magnitude in comparison to Cs clocks. Previous efforts to measure this effect in Ca, however, were hampered by low sample densities n ~ 10^9 cm-3, resulting in measured shift coefficients with errors exceeding their associated absolute values. To overcome this limitation, we have performed absolute frequency measurements with 1 microKelvin 88Sr atoms for densities spanning three orders of magnitude, from 10^9 cm-3 to 10^12 cm-3. For the first time we report definitive density-related line center shifts and spectral broadening of an optical clock transition in ultracold atoms. We find the density-dependent fractional frequency shift is 250 times smaller than that of Cs.

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